



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2024

INFORMATION AND COMMUNICATIONS TECHNOLOGY (N540)

Candidates answer on the Question Paper.

TIME: 1 HOUR 45 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Any rough working should be done in this booklet.
5. Do NOT use calculators.
6. Answer **ALL** questions.
7. This document consists of **11** questions printed on **25** pages, numbered **2** to **26**.
8. The number of marks is given in brackets [] at the end of each question or part question.
9. The total marks for this paper is **100**.

For Examiners' use													
Question No.	1	2	3	4	5	6	7	8	9	10	11	Total	Signature
Examiner													
Team Leader													
CE/ACE													

SECTION A

Question 1 (15 marks)

Circle the correct answer. Each item carries one mark.

(a) Which one of the following is an **output** device?

A



Touchpad

B



Plotter

C



Microphone

D



Joystick

(b) How many bits make up a **byte**?

A 16 bits

B 8 bits

C 12 bits

D 4 bits

(c) Which component prevents the CPU from overheating?

A Fan

B Sound card

C Graphics card

D Motherboard

(d) Which one of the following **stores** data temporarily?

A



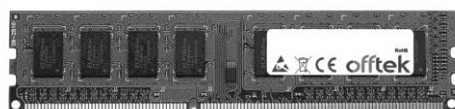
ROM

B



Hard Disk

C



RAM

D



CD ROM

(e) Which **device** is shown in the picture below?



- A** Digital camera
- B** Barcode reader
- C** Headphone
- D** Printer

(f) The data below shows the **marks** obtained by Ravi in a test.

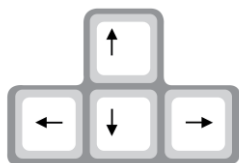
	A	B	C	D
1	Subject	Marks		
2	English	70		
3	French	85		
4	Maths	69		
5	Science	90		
6				

In which cell do the **marks** for **French** appear?

- A** A2
- B** B2
- C** A3
- D** B3

- (g) In a PowerPoint presentation, which **icon** illustrates “starting the slide show from the **first slide**”?

A



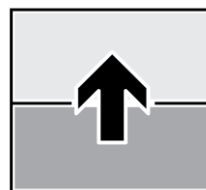
B



C



D



- (h) Which **speech balloon** is used when you want your character to tell a secret?

A



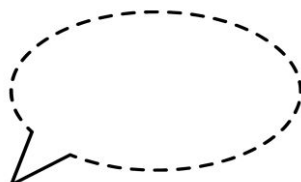
Sound Balloon

B



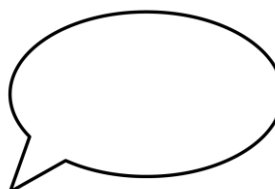
Thought Balloon

C



Whisper Balloon

D



General Speech Balloon

(i) Which **component** is used to connect two **LANs**?

A



Switch

B



Server

C



Router

D



Cable

(j) Which **hardware component** is shown below?



A

Internal Memory

B

Hard-disk

C

Network Interface Card

D

Wireless Fidelity Card

- (k) Which health problem may be caused when a **mouse** is used for a long time?



- | | | | |
|----------|--------------------------|----------|-----------|
| A | Repetitive strain injury | B | Headache |
| C | Eye strain | D | Back pain |

- (l) Which of the following is a popular **videoconferencing** software?



Twitter



Wiki



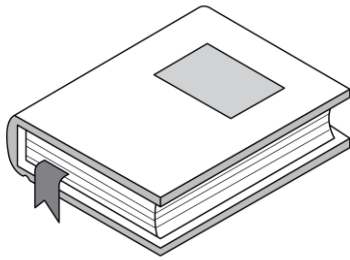
Facebook



Skype

(m) Which one of the following is required for **e-learning**?

A



Book

B



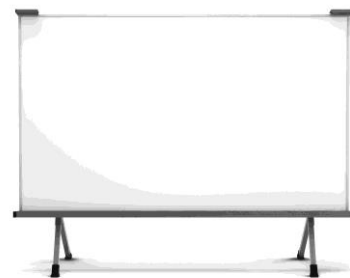
Calculator

C



Laptop

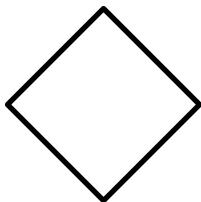
D



Whiteboard

(n) Which **one** of the following is the symbol for “**Decision Box**” in a flowchart?

A



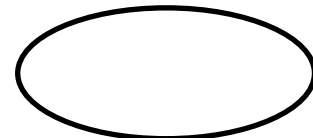
B



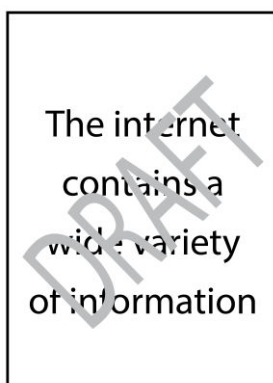
C



D



- (o) Which feature in MS Word allows the word “**DRAFT**” to be inserted in a document, as shown below?



A Page Borders

B Watermark

C Ruler

D Page Color

E

TL

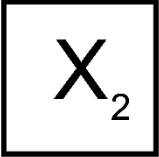
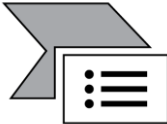
CE

Question 2 (10 marks)

- (a) Label the following **icons** using the terms given below.

There is an extra word in the list.

AutoSum	Font Color	Strikethrough
Subscript	SmartArt	Superscript

[5]

- (b) Fill in the blanks with the **correct term** from the given list.

There is an extra word in the list.

backup	bus	mesh
virus	Linux	Wifi

- (i) In a topology every node is connected to a single cable.
- (ii) An example of a free operating system is
- (iii) Mobile phones can be connected to the internet using
- (iv) A is a software that can slow down a computer.
- (v) The purpose of a is to keep data secure.

[5]

E

TL

CE

Question 3 (10 marks)

Tick (✓) **True** or **False** next to each of the statements below. An example is given.

		True	False
a)	A mouse is an example of an input device.	✓	
b)	A laser printer is slower than an inkjet printer.		
c)	Page break helps to separate content between pages.		
d)	A conditional formatting rule cannot be removed in Excel.		
e)	In a PowerPoint Presentation, you cannot use more than one slide master.		
f)	In a star topology, nodes are connected to a central computer.		
g)	A firewall can be a hardware or a software.		
h)	Personal information should be shared with everybody.		
i)	In a flowchart, flowlines indicate the movement of data.		
j)	The report wizard is located in the Home Tab in MS Access.		
k)	Python is an example of a database software.		

[10]

Question 4 (10 marks)

(a) Six hardware devices are shown below.

 Projector	 3D Printer
 Touchscreen	 Keyboard
 Solid State Drive (SSD)	 Memory Card

Choose a suitable **hardware device** from the list above that can be used to perform the following tasks:

An example is given.

(i)	Rapidly enters data in public places	Touchscreen
(ii)	Stores large volumes of data in smartphones	
(iii)	Stores pictures in digital cameras	
(iv)	Creates real world objects	
(v)	Magnifies images generated by a computer	

[4]

- (b) Match each term in **Column A** to the correct statement in **Column B**.

Write your answers in the **answer grid** provided below.

An example is given.

There is an extra statement in **Column B**.

Column A	
(i)	Mailing list
(ii)	Memory Management
(iii)	Hacking
(iv)	Scanner
(v)	Blog
(vi)	Google
(vii)	Chat

Column B	
1	can produce digital images of hardcopy documents.
2	is an example of a search engine.
3	is an example of text-based communication.
4	is an online journal that displays the latest posts first.
5	is used to perform mail merge in a word document.
6	deals with the loading and running of application software.
7	is an example of an online forum.
8	refers to gaining unauthorised access to a computer system.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
5						

[6]

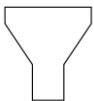
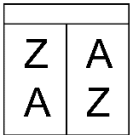
E
TL
CE

Question 5 (5 marks)

(a) Using the **terms** given below, name the following **icons**.

An example is given.

Format Painter	Sort	Slide Master	Filter	Notes Page
-----------------------	-------------	---------------------	---------------	-------------------

	Filter
	
	
	

[3]

(b) The following statements concern **Data Security**.

Tick (✓) **two threats** to data security.

Using strong passwords ☐

Accidental loss of data by deletion ☐

Data destroyed by flood ☐

Using an updated anti-virus ☐

[2]

SECTION B**Question 6 (5 marks)**

The specifications of a computer system are shown below.

View basic information about your computer

Windows edition _____

Windows 10 Home Single Language

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System _____

Processor: Intel(R) Core (TM) i5-7200U CPU @ 2.50GHz

Installed memory (RAM): 8.00 GB (7.87 GB usable)

System type: 64-bit Operating System, x64-based processor

Pen and Touch: No pen or Touch Input is available for this Display

Computer name, domain, and workgroup settings _____

Computer name: laptop013

Computer description:

Workgroup: WORKGROUP

Windows activation _____

Windows is activated [Read the Microsoft Software License Terms](#)

Product ID: 00327-30523-29858-AAOEM

(a) What is the **name** of the computer?

.....

[1]

(b) Tick (✓) the correct **memory size** of the computer.

8 GB ☐

16 GB ☐

32 GB ☐

[1]

(c) What is the **model** and **speed** of the processor used in the computer?

Model

[1]

Speed

[1]

(d) Give the name of the **operating system** installed on the computer.

.....

[1]

E
TL
CE

Question 7 (6 marks)

- (a) Reshma has to insert a **Table of Contents** in her project work.

Write **2**, **3** and **4** in the column '**Order**' to indicate the order of the **steps** to be taken to insert the **Table of Contents**.

The first one has been done for you.

Step	Order
Click Table of Contents on the Reference Tab.	
Choose one of the types of Table of Contents available.	
Format document using Heading Style found on Home Tab.	1
Place cursor where you want the Table of Contents to appear.	

[3]

- (b) A letter is to be sent to a group of people. **Figure 1** shows the letter with the merged fields.

12 May 2024

<<Address>>

<<Town>>

SALSA DANCE GROUP

Dear <<First Name>> <<Surname>>

You are invited to the opening of our dance club on Sunday 30 June 2024 at 19 00.

You will perform in the category <<Category>> on that day.

Yours faithfully

Chaya

Secretary

Figure 1: Letter with merged fields

Table 1 shows the list of mail merge recipients.

Table 1: List of Mail Merge Recipients

	Surname	First Name	Category	Address	Town
	Pierre	Laura	Ballet	Royal Road	Curepipe
✓	Khan	Parwez	Fusion	Branch Road	Rose Hill
	Ram	Shreya	Bollywood	Hillcrest Road	Black River

Complete the letter below for the recipient indicated by the tick (✓) in **Table 1** to show the result of the mail merge.

12 May 2024

.....

.....

SALSA DANCE GROUP

Dear

You are invited to the opening of our dance club on Sunday 30 June 2024 at 19 00.

You will perform in the category..... on that day.

Yours faithfully

Chaya

Secretary

[3]

Question 8 (10 marks)

The spreadsheet below shows the data on **sales of perfume bottles (\$)** in a duty-free shop.

Refer to the spreadsheet and answer the questions that follow.

	A	B	C	D	E	F
1	Brand	January	February	March	Average Sales	Below Average
2		\$	\$	\$	\$	
3	Dior	2700	3500	4000	3400	Above
4	Chanel	2000	2200	2100	2100	Below
5	Yves Saint Laurent	2500	3000	3200	2900	Above
6	Gucci	1800	1900	2000	1900	Below
7	Dolce	2300	1700	2600	2200	Below
8	TOTAL SALES	11300	12300	13900		

(a) Write down the **cell address** which contains the following information:

- (i) **Sales of Chanel** perfume bottles in January:
- (ii) **Average sales of Gucci** perfume bottles:

[2]

(b) Tick (✓) the correct formula that has been used in cell **C8** to calculate the **Total Sales** for the month of **February**.

= C3 * C7 ☐ = SUM (C3:C7) ☐ = C3 – B7 ☐

[1]

(c) Write the **formula** that can be used in cell **E3** to find the **Average Sales** of **Dior** perfume bottles.

.....

[2]

- (d) The **Shop Average Sales = \$ 2500**.

When the Average Sales of a perfume bottle is **less than** the Shop Average Sales, **Below** should be displayed.

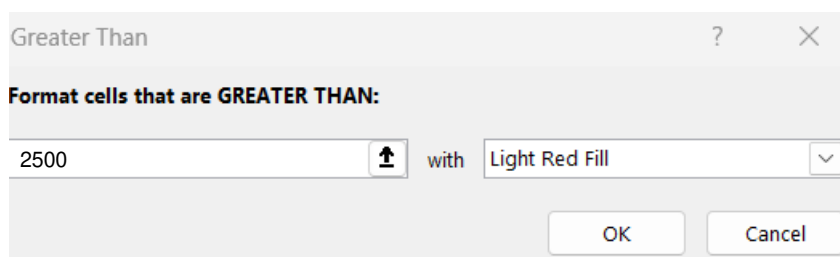
Otherwise **Above** should be displayed.

Complete the following formula to be used in **cell F3**.

= IF (E3 <, "Below",)

[2]

- (e) Conditional formatting is applied to cell E3:E7 as follows:



Name the **cells** that will be formatted.

.....

[2]

- (f) The formula **= COUNTIF(F3:F7,"Above")** is used in cell **F8**.

What will be the output?

.....

[1]

E
TL
CE

Question 9 (9 marks)

- (a) Match the term in **Column 1** to the correct statement in **Column 2**.

Column 1		Column 2
Data type	•	• Is used to organise and display data.
Report	•	• A group of records.
		• Can be added to a specific field.

[2]

- (b) A database table **GAMES** was set up as shown below.

GAME ID	TITLE	RATING	TYPE	PRICE (€)
Z12	BOMB JACK	5	Platform	50
C34	GALAXIAN	4	Shoot	65
B76	TEMPLE TERROR	5	Adventure	60
A45	HERO QUEST	3	Fighting	80
Q64	MANIC MINER	4	Platform	30

- (i) Which **data type** is used for the field **RATING**?

.....

[1]

(ii) A **query by example** grid is shown below.

Field	GAME ID	RATING
Table	GAMES	GAMES
Sort	ASCENDING	
Show	☒	☐
Criteria		>4
Or		

Show what the **output** would be.

.....

.....

.....

[3]

(iii) Complete the **query by example** grid below to select and show the **name of games** of **type** Platform.

Field		
Table	GAMES	GAMES
Sort		
Show	☐	☐
Criteria		
Or		

[3]

E
TL
CE

Question 10 (9 marks)

Internet banking is becoming very popular nowadays.



- (a) Explain what is meant by **internet banking**.

.....
 [2]

- (b) State **two** transactions that can be done using internet banking.

1.
 2. [2]

- (c) Banks usually provide E-banking websites to their customers.

State **two** pieces of information that customers will need in order to access their accounts through an E-banking website.

1.
 2. [2]

- (d) Give **two** benefits of internet banking for customers.

1.
 2. [2]

- (e) State **one** possible drawback of internet banking for a bank.

..... [1]

Question 11 (11 marks)

- (a) Fill in the blanks using the **terms** given below.

There is an extra term in the list.

flowline	sequence	loop	flowchart
-----------------	-----------------	-------------	------------------

- (i) A is a diagram that shows the steps to solve a problem.
- (ii) A connects one symbol to another.
- (iii) Steps in a are executed one after the other from first to the last.

[3]

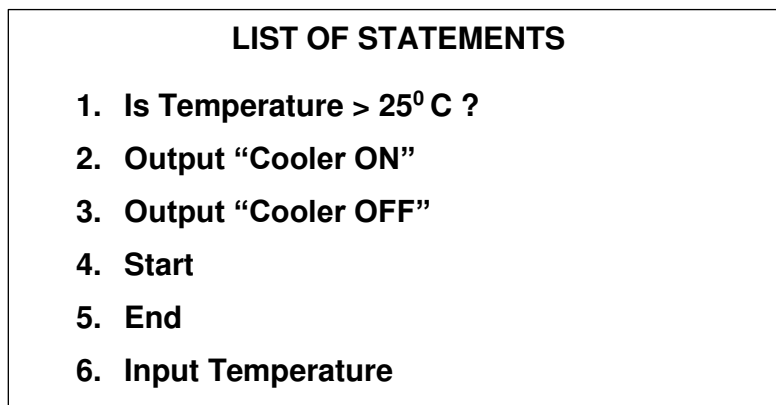
- (b) Complete the **Python program** to display the sentence “The weather is windy.” **seven times**.

```
>>> count = 0
>>> while count < ..... :
    Print ( ' ..... ' )
    count = ..... + .....
```

[3]

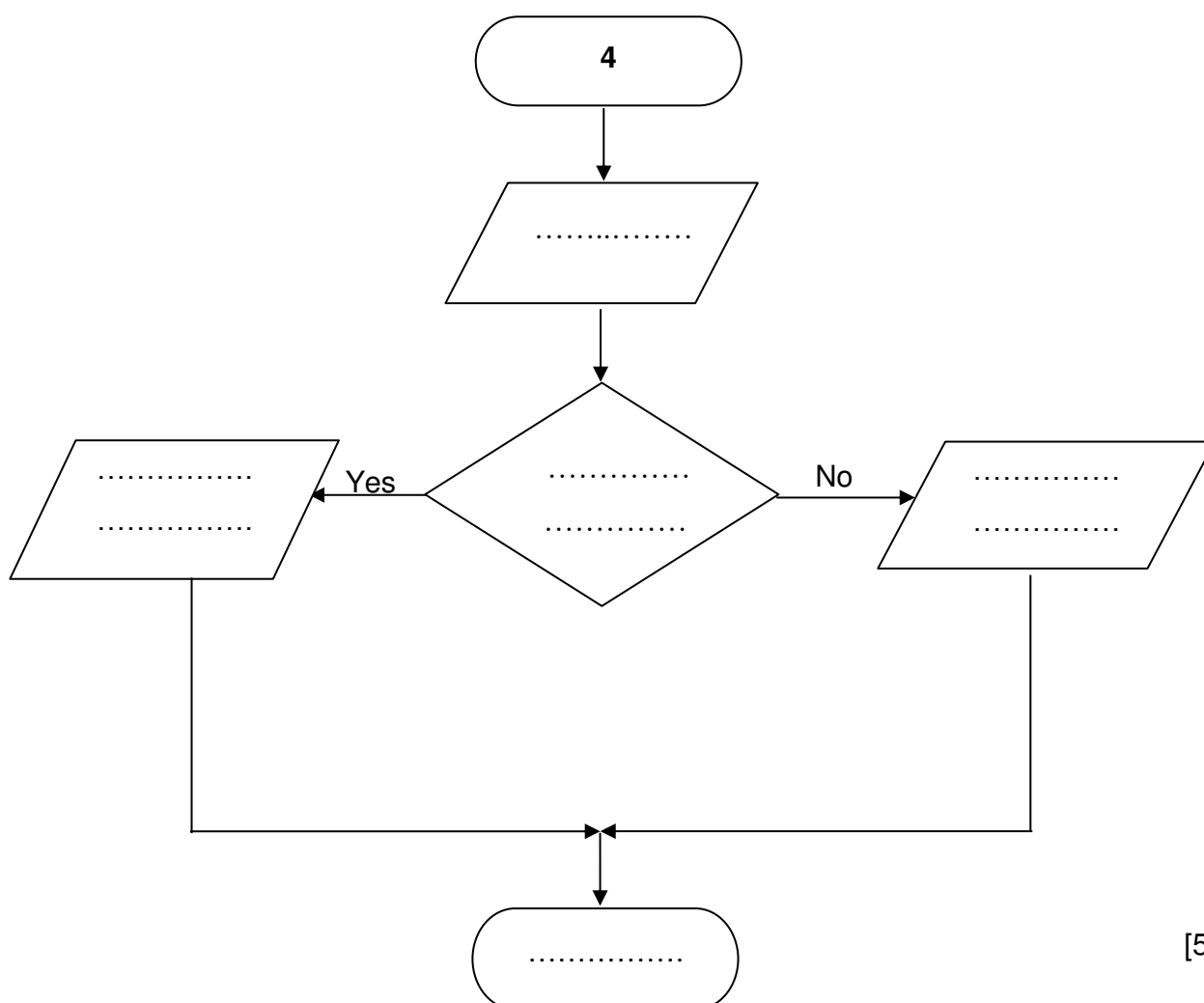
- (c) An algorithm has to be designed for a sensor using a flowchart. The aim of the sensor is to detect if the temperature in a greenhouse is greater than 25° C, in order to start a cooler.

A list of statements is provided below.



Using the list of statements provided above, insert the **six missing statements** in the flowchart using the appropriate **number**. Each statement may be used only once.

An example is given



[5]

BLANK PAGE

BLANK PAGE